An 8-bit 20.7 TOPS/W Multilevel Cell ReRAM Macro With ADC-Assisted Bit-Serial Processing

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***Abstract*— Analog compute in memory (CIM) with multilevel cell (MLC) resistive random access memory (ReRAM) promises highly dense and efficient compute support for machine learning and scientific computing. This article introduces analog to digital converter (ADC)-assisted bit-serial processing for efficient, high-throughput compute. Bit-serial digital to analog converters (DACs) and 8-bit binary-weighted multicycle sampling (BWMCS) ADCs perform analog vector-matrix multiplication (VMM) on MLC-based crossbar arrays. A direct drive g_m-boosted transimpedance amplifier (TIA) enables high-speed crossbar readout. We present a system on chip (SoC) prototype consisting of four self-contained ReRAM-based CIM macros and a reduced instruction set computer-five (RISC-V) host. The test chip is fabricated in 65 nm CMOS with foundry-integrated MLC ReRAM. We trained LeNet1 for handwritten digit classification and mapped the CNN weights differentially to 3-bit MLC ReRAM across multiple CIM macros. The classification accuracy loss is 1.6% when compared to the quantization-aware trained model. The measured raw and normalized peak efficiencies are 20.7 and 662 TOPS/W, respectively. The compute density is 8.4 TOPS/mm^2.**

***Index Terms*— Analog to digital converter (ADC), bit-serial, compute-in-memory (CIM), resistive random access memory (ReRAM).**

I. Introduction

IN THE rapidly evolving landscape of computing technologies, analog compute in memory (CIM) with multilevel cell (MLC) resistive random access memory (ReRAM) shows promise toward meeting the demands of machine learning and scientific computing. The allure of analog CIM lies in its potential to break the Von Neumann memory wall, where data transfer between memory and processing units often bottlenecks performance. Executing compute operations directly within a nonvolatile memory (NVM) array with mixed-signal circuits reduces data movement and improves latency. Multibit ReRAM is a highly dense NVM technology that maximizes pre-analog to digital converter (ADC) information for high-efficiency compute [1]. Careful peripheral mixed-signal circuit design, massive parallelization, and the utilization of multibit NVM improve energy efficiency and compute density. Collectively, by maximizing resource utilization across the entire array, energy consumption is minimized.

Quantized neural networks (QNNs) leverage integer-only arithmetic during training and inference to reduce latency and overall model size while maintaining near-identical floating-point accuracy. Mapping digital QNN vector-matrix operations to analog ReRAM crossbar arrays significantly improves computational speed and energy efficiency.

High-precision input and output, as well as NVM cells, are essential for practical analog CIM processing elements (PEs). Floating-point accuracy is possible with both sufficient bit-width and neural network training techniques. Post-training quantization (PTQ) is effective for neural network weight compression down to 8 bits but is ineffective for lower weight bit-widths and for low precision input and output. PTQ neural network accuracy suffers from aggressive quantization and low bit width representation. Quantization-aware training (QAT) is a technique that utilizes input, output, and weight bit-widths down to 2 bits for simple neural networks. However, modern computing workloads leveraging QAT demand at least 4-bit weight quantization and 8-bit input/output quantization [2], [3].

Recent design optimization of high-precision crossbar array interface circuits has led to many interesting architectures. Typically, the precision of the PE drives the overall architecture and associated challenges. Most published analog CIM works implement 1-bit ReRAM in the PE [4], [5], [6], [7], [8], [9], [10], [11], [12]. For higher bit-width representation, the common approach is to spread 1-bit ReRAM cells across multiple columns on the same row. Weight spreading requires digital accumulation and shift-add circuits, which incur a

Received 4 June 2024; revised 23 September 2024 and 12 January 2025; accepted 2 February 2025. Date of publication 21 February 2025; date of current version 25 July 2025. This article was approved by Associate Editor Meng-Fan Chang. This work was supported in part by the Applied Materials. *(Corresponding author: Justin M. Correll.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/JSSC.2025.3540114.

Digital Object Identifier 10.1109/JSSC.2025.3540114

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latency cost for full 8-bit weight operation [4], [5], [7], [12]. Dedicated column circuitry helps, but overall compute density suffers due to weight spreading. Multibit ReRAM operation is first demonstrated in [13] with a three-level (ternary) weight representation and on-chip write circuitry. However, the design has only one low-precision ADC and does not use pitch-matching, and therefore is not optimized for future many-bits-per-cell ReRAM programming [14], [15] or overall resource amortization. Upton et al. [16] demonstrate a multiple-bits-per-cell ReRAM macro up to 16 levels, but the ReRAM cell conductance range is substantial (0–256 μs) and not suitable for large row activations since column current accumulation would be extreme.

Recent works implement 1–4 bit ADCs to digitize column currents [6], [8], [9], [10], [13]. Moreover, said designs deploy flash ADC converters to leverage their simple structure and speed advantage. However, flash converters are unsuitable for high-precision compute since each additional bit of precision doubles the number of comparators. As the ADC area increases, achieving effective pitch matching becomes more challenging. This leads to reduced circuit parallelization, which negatively impacts CIM macro performance. In [8], the columns of a 256 $\times$ 256 binary crossbar array terminate into eight 1–4 bit flash ADCs. A single ADC is shared among 32 columns, which illustrates the pitch-matching challenge. In [13], the 4-bit flash ADC is used for both column readout and for iterative multilevel ReRAM write-verify cell programming. Sharing the ADC is prudent but slows down the array. Future applications require more precision, which is a challenge for the flash ADC architecture.

Building upon these observations, we summarize the numerous hardware co-design challenges that persist toward the optimization of analog CIM architectures as follows: 1) leveraging mixed-signal circuits for high-precision compute with low latency; 2) designing peripheral array circuits that maximize parallelization for low-voltage MAC operations and that meet the high-voltage requirements essential for accurate multibit ReRAM programming; 3) effectively utilizing on-chip multibit ReRAM for increased compute density; and 4) managing data throughput and latency from host CPU to analog CIM macros.

To address the challenges summarized in Table I, we introduce ADC-assisted bit-serial processing for high throughput, high accuracy, and low-power vector-matrix multiplication (VMM). Bitwise or bit-encoded input is presented in [4], [5], and [7], but wastes clock cycles due to long bitline (BL) charge time and subsequent discharge-based readout. In our approach, an 8-bit VMM is computed in eight clock cycles at an ADC efficiency of 7.8 pJ per 8-bit MAC operation. Pitch-matched driveline (DL) 1-bit digital to analog converters (DACs) apply eight binary encoded bitwise pulses to crossbar rows. The BL columns are terminated with transimpedance amplifiers (TIAs) that directly drive 8-bit binary-weighted multicycle sampling (BWMCS) ADCs. A critical contribution of this work is that the BWMCS ADC first efficiently scales the eight partial bitwise outputs and then asynchronously digitizes the overall array-level MAC result.

Pitch-matching the TIA and ADC footprints to the crossbar column pitch is challenging due to the small column pitch of 1.92 μm set by the 1.92 $\times$ 1.22 μm 1T1R bitcell. Our compact TIA and BWMCS terminate a pair of BL columns. Furthermore, we leverage a multibit dual-array neural-network mapping technique with positive and negative weights [17] to cancel any channel-dependent nonidealities such as channel offset and ADC offset [11].

TABLE I
ANALOG CIM CHALLENGES

Challenge	Summary
Throughput	PWM is slow and non-linear
Efficiency, Pitch matched circuitry	Mixed-signal circuits consume too much energy and occupy too much area
Multi-level ReRAM	Not commercially available
Compute Density	Bulky mixed-signal circuits and high-voltage circuits

Peripheral array circuits must support both low-voltage VMM and high-voltage multibit ReRAM programming, which presents a challenge for pitch-matching. High-voltage multibit ReRAM write-verify programming requires bulky high-voltage transistors, level shifters, and pass gates. Our pitch-matched switch architecture limits the number of large high-voltage transistors and critically minimizes the number of level shifters in each row and column. Each DL row contains a 1-bit read DAC, DAC-enable switch, ground (GND) switch, and local WRITE-enable switch. The WRITE-enable level shifter is the only one within each DL switch circuit block. The DAC-enable switch protects the read DAC from high voltage and is driven by a global level shifter. Each BL column contains a TIA-enable switch, GND switch, and WRITE-enable switch—all with associated level shifters. In the case of the BL, the level shifters fit nicely within the TIA/ADC footprint.

Multibit ReRAM is the holy grail of analog CIM but demands extensive support circuitry for device characterization and usage with CNN parameters. Multilevel ReRAM requires accurate high-voltage programming, low-voltage high ADC precision readout, and iterative write-verify programming algorithms [13], [16], [18], [19]. We leverage a TIA high gain mode, the BWMCS ADC, and our reduced instruction set computer-five (RISC-V) processor for write-verify iterative programming to facilitate 4-bit ReRAM programming with 16 levels below 4 μA, which improves upon [16] by 16$\times$.

Data movement in and out of the crossbar macro is critical for efficient analog CIM. We use an on-chip RISC-V host for CNN inference workload management. A modified AXI direct memory access (DMA) controller supports 2.0 Gb/s burst mode data transfer between host and macro or between macros. Decoupled workload execution achieves higher efficiency, thanks to a local macro controller that offloads MAC operations from the host. Local compute is executed autonomously from the host.

This article expands on the conference paper [20], and the remainder is organized as follows. Section II covers the crossbar architecture, including macro design, operating modes, bitwise MAC operation, and switch configurations. Section III details the design of the ADC, TIA, and word

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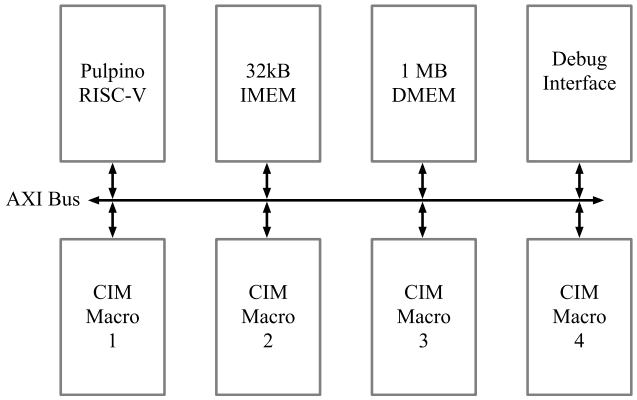


Fig. 1. RISC-V SoC architecture with four ReRAM-based CIM macros.

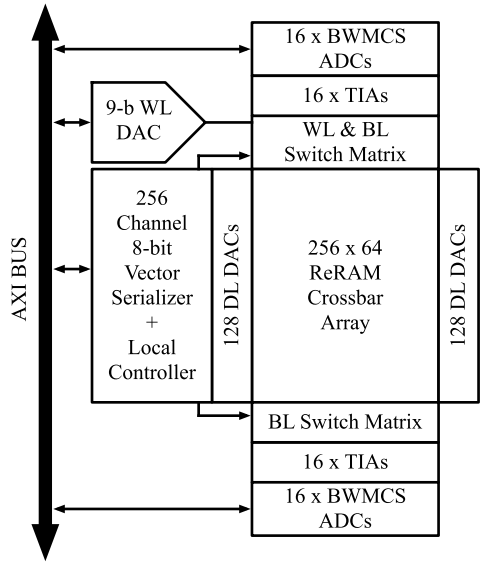


Fig. 2. Block diagram of the self-contained CIM macro.

line (WL) DAC. Section IV discusses system-level control via the RISC-V and local MAC controller. Section V presents our prototype and measurement results. Finally, Section VI is the conclusion.

II. Crossbar Architecture

A. Crossbar Macro

The RISC-V system on chip (SoC) prototype comprises four ReRAM-based CIM macros that support instruction-programmable end-to-end DNN inference (Fig. 1). Fig. 2 shows one of the four self-contained CIM macros in the prototype. The 256 × 64 crossbar array uses foundry 1.92 × 1.22 μm 1T1R filamentary ReRAM bitcells with a 3 V NMOS selector as the NVM cells. The crossbar macro includes all the DACs, ADCs, and switches required to support formation, write verification, and MAC operation. The RISC processor loads macro registers to configure the mixed-signal circuits and the switch matrices that interface to the array. Fast DMA transfers crossbar input and output data between memory-mapped registers on the crossbar and the CPU. A local controller directs the MAC operations and alerts the CPU on MAC completion, significantly reducing the load on the CPU.

TABLE II
Operating Modes

Mode	Voltage	Pulse Duration	# Pulses
FORM	3 - 4 *V*	100 *ns*	3
SET	4 *V*	100 *ns*	1
RESET	2 *V*	100 *ns*	1
WRITE-VERIFY	0 - 2 *V (write)* 300 *mV (verify)*	100 *ns* *32ns*	50 *write* 50 *verify*
VMM EVEN	300 *mV*	32 *ns*	8 *RTZ*
VMM ODD	300 *mV*	32 *ns*	8 *RTZ*

Pitch-matched mixed-signal circuits support high compute density, high throughput, and low power. 256 DACs apply the 8-bit inputs to the DLs. 32 8-bit ADCs and TIAs support 64 crossbar array BLs and connect through the BL switch matrix. Positive and negative weights are implemented on adjacent columns (i.e., even and odd column numbers) which share the same ADC and TIA, thus gain and offset errors are common modes and easily canceled. A 9-bit WL DAC provides the gate voltage for the 1T1R bitcells. The WL switch matrix connects the 9-bit DAC to the even or odd columns during MAC or to a single column for writing and forming. The DL, BL, and WL switch matrices configure the array for MAC, form, set, reset, and single device measurement.

Dedicated row and column hardware supports a diverse set of array operations. Write hardware selects individual ReRAM bitcells for high-voltage, bidirectional potentiation. The write circuitry provides current compliance during ReRAM filament formation and fine-grained tuning during programming. Read hardware performs individual bitcell write-verify as well as parallel VMM at low voltage. Specialized analog switch matrices enable high-voltage operations such as writing and forming and protect the low-voltage read hardware.

B. Operating Modes

Table II shows the six different operating modes of this prototype SoC. The first four modes support high-voltage ReRAM bitcell programming and the remaining two support low-voltage VMM partial-sum compute. The host processor directs the bitcell programming algorithms, initiates VMM partial summation, performs activation functions, and controls the data path. Local macro configuration registers on the CIM macro store the mode settings.

Fig. 3 shows the crossbar configurations for FORM and RESET of a single bitcell. In both cases, the WL DAC biases the 1T1R gates of a single column. All other WL columns are grounded, thereby switching off unselected bitcells. The selector transistors of the unselected columns are disabled, providing high impedance, and inhibiting any erroneous current flow. In this way, the 1T1R selector transistor provides isolation and inhibits sneak-path currents in neighboring bitcells.

During FORM, a single DL row connects to the VFORM voltage while all other DLs are grounded. All the BLs connect to the ground. Current flows from the VFORM voltage through the selected device to ground. The VFORM voltage is applied only across the selected device and kept within the 4.6 V

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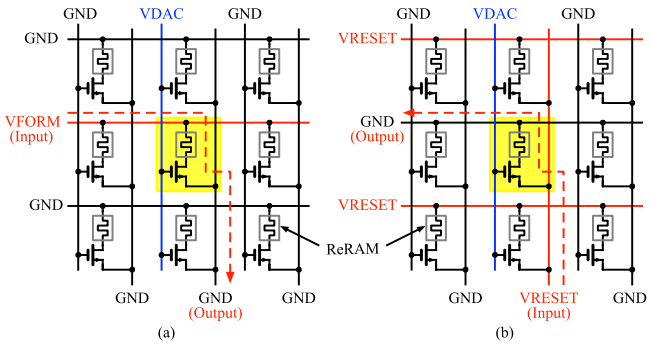


Fig. 3. (a) ReRAM FORM and (b) RESET.

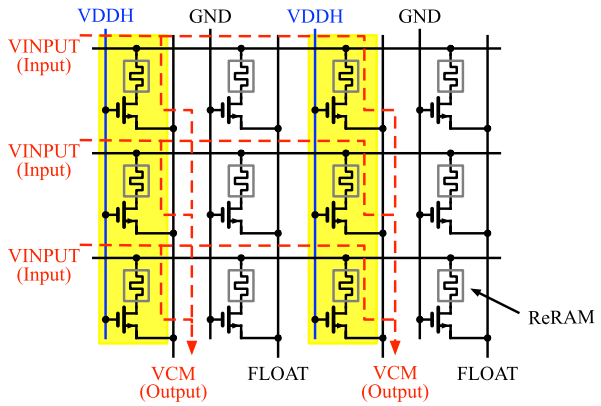


Fig. 4. MAC read, even mode.

maximum voltage rating of the selector transistor to ensure reliability.

In RESET mode, a single BL connects to VRESET while all other BLs connect to the ground. At the same time, a single DL is grounded, while all other DLs are biased by VRESET. In this way, current flows from the VRESET voltage on the BL through the selected bitcell to the grounded DL.

During read operations, the array operates in two phases to compute VMM partial sums utilizing positive and negative matrix coefficients, respectively. In the first phase, row read DACs are loaded with input data and connected to the crossbar. Even columns connect to the 32 TIA inputs, each providing a 900 mV common-mode virtual ground to the column. The Read DACs serialize the 8-bit input activation vectors into 900 mV to 1.2 V return to zero (RTZ) pulse trains. The RTZ pulses linearize the bit-cell operation thanks to the 1-bit DAC and TIA virtual ground [18]. Crossbar rows are excited with fixed 300 mV pulses (which do not disturb the ReRAM state), giving rise to bitcell currents that sum along the columns for a MAC operation. Column TIAs convert these MAC currents to voltages for digitization by 8-bit ADCs, resulting in the VMM partial sum output activation vector. Fig. 4 shows an example of the even phase. In the second phase, the process repeats on the odd columns utilizing the same input activation vector. Both ADC output activations for VMM partial sums are subtracted and then stored by the processor.

C. Bitwise MAC Operation

We introduce ADC-assisted bitwise crossbar operation for high throughput, high accuracy, and low power. PWM or

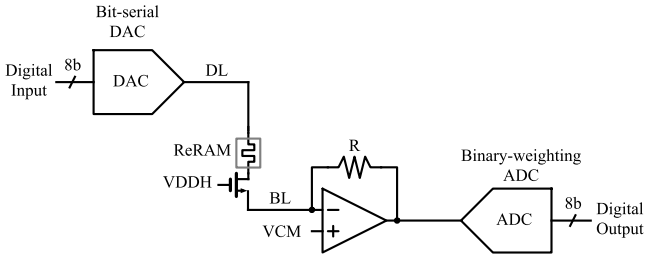


Fig. 5. Bitwise operation.

pulsed DAC operation overcomes bit-cell nonlinearity but suffers from slow throughput due to the requirement for 2^N time steps [18], [19]. Moreover, PWM operation involves integration in the BL or mixed-signal circuity. Bitwise operation requires only N crossbar operations for an N-bit input. However, digitizing, combining and weighting can be energy intensive, especially if separate intermediate ADC results must be weighted and added in the digital domain.

In our bitwise approach (Fig. 5), the DL DACs apply N single-bit pulses corresponding to the N bits of the digital input. A TIA converts the BL output current to voltage pulses and drives the ADC input. The ADC weights the input before performing a single digitization of the overall MAC output. A critical advantage is there is only a single ADC digitization for each MAC operation, greatly improving speed and energy efficiency. Moreover, as we see in Section III, we reuse the ADC capacitances to efficiently weight the N voltage pulses.

D. Switch Configurations

For optimum compute density, the mixed-signal support circuitry fits the 1.22 μm row and 1.92 μm column pitch of the array. Row circuitry is placed on both sides of the crossbar, resulting in an effective row circuit pitch of 2.44 μm. Similarly, the column support circuitry is located at the top and bottom of the crossbar array. With each ADC and TIA supporting two columns, the resulting circuit column pitch is 7.68 μm.

This tight pitch is especially challenging for the switches, which must support low voltages for reading and high voltages for programming. High voltages not only necessitate high-voltage transistors but also require level shifters to translate the digital control signals to the high-voltage gate signals. High-voltage transistors are considerably larger than logic devices. Moreover, level shifters require multiple high-voltage FETs and are particularly large. Our approach limits the number of large high-voltage transistors and critically minimizes the number of level shifters.

Fig. 6 illustrates the switch strategy, focusing on the row (i.e., DL) circuitry. During MAC read, a single-bit DAC applies N single-bit pulses to the DL. The single-bit DAC uses low-voltage transistors, S_a and S_b, to switch between the read voltage (VREAD) and the common mode voltage (VCM). The write circuity applies VWRITE (up to 4 V) or 0 V to the DL to support bitcell form or write. A high-voltage transistor switch, S_c, isolates the DAC from high voltages during these high-voltage operations. This NMOS switch shares the same

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TABLE III
MACRO AREA OVERHEAD

Macro Block	# of Blocks	Location	Width [μm]	Height [μm]	Area [mm^2]
Row DACs	128	Left	100	317	0.0317
Row DACs	128	Right	100	317	0.0317
WL MUX	16	Top	138	35	0.0048
BL MUX	16	Top	138	39	0.0054
TIA	16	Top	146	41	0.0060
ADC	16	Top	146	210	0.0307
BL MUX	16	Bottom	138	39	0.0054
TIA	16	Bottom	146	41	0.0060
ADC	16	Bottom	146	210	0.0307
256x64 Crossbar	1	Center	135	322	0.0435
WL DAC	1	Top Left	94	222	0.0209
				Total	**0.2168**

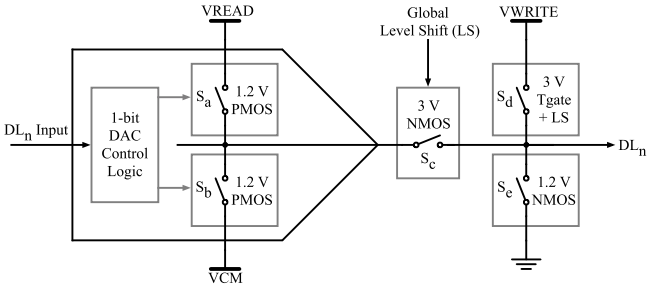


Fig. 6. Row circuitry showing switch type and level shifters.

gate control for the entire array, and, therefore, can also share a single global level shifter. Switches S_d and S_e connect the DL to VWRITE or ground. Switch S_d is a transmission gate with an associated level shifter. Although switch S_e, which connects to the ground, is a high-voltage NMOS transistor, it operates well with a 1.2 V digital gate signal. Table III summarizes the area overhead of each of the circuit blocks.

III. CIRCUIT BLOCKS

A. Binary-Weighting Multicycle Sampling ADC

The ADC not only digitizes the column outputs but also weights column outputs to support the bitwise operation. As mentioned in Section II, the MAC operation is decomposed into multiple bitwise operations. A binary-weighted summation produces the final analog result before quantization. Fig. 7 illustrates the functional operation of the binary-weighted summation and the quantization (i.e., the MAC backend).

A sampling clock (CLK$_S$) and a conversion clock (CLK$_C$) control the MAC backend. At the falling edge of CLK$_S$, a sampler (S/H) samples the output from the MAC array (V_{IN}), adds this to the previous sampled value, and then halves the total sampled value. At the rising edge of CLK$_C$, the quantizer quantizes the final sampled value and outputs a corresponding digital code (D_{OUT}). Meanwhile, the sampled value is also cleared by CLK$_C$.

CLK$_S$ occurs successively several times before CLK$_C$ occurs, such that a V_{IN} series is sampled and summed with

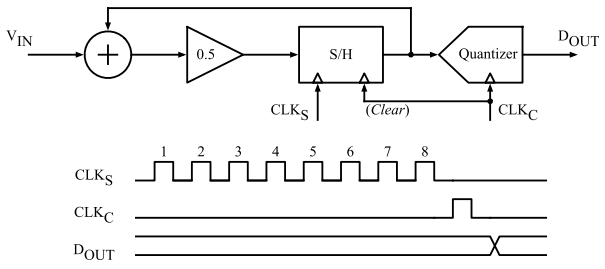


Fig. 7. Behavioral model of the MAC backend and timing diagram.

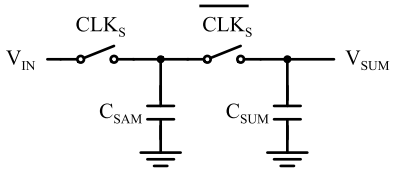


Fig. 8. Passive binary-weighted sampler.

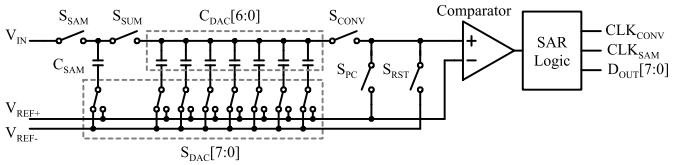


Fig. 9. BWMCS ADC.

binary weighting. Equivalently, every V_{IN} is summed with a weight of 2^{-n}, where n is the reverse order of sampling before CLK$_C$. The overall function of the MAC backend can be described by this equation

$$D_{OUT} = \left\lfloor \frac{2^{-1}A_{IN}[n] + 2^{-2}A_{IN}[n-1] + \cdots + 2^{-n}A_{IN}[1]}{V_{fullscale}} \times 256 \right\rfloor. \quad (1)$$

Note that, the first sample of V_{IN} gets the smallest weighting, corresponding to an LSB-first MAC order.

A straightforward implementation of Fig. 7 (excluding quantizer) consisting of an active adder and a divider would require significant power and area. In this work, we introduce simple passive switched-capacitor circuitry to implement the sampler, adder, and divider simultaneously, as shown in Fig. 8.

The proposed sampler consists of two capacitors and two switches. When CLK$_S$ goes high, a sampling capacitor (C_{SAM}) tracks the input signal. Next, when CLK$_S$ goes low, V_{IN} is held on C_{SAM}, and C_{SAM} is then charge shared with a summing capacitor (C_{SUM}). To implement binary weighting, C_{SAM} is equal to C_{SUM}, so that after every charge sharing ($\overline{\text{CLK}_S}$), $V_{SUM} = 0.5(V_{SUM}z^{-1} + V_{IN})$.

We employ an SAR ADC architecture for compactness and high efficiency and employ the sampling scheme of Fig. 8. This 8-bit SAR ADC operates in a single-ended, and top-plate sampling configuration. The SAR uses an efficient strong-ARM comparator with a preamplifier to reduce kick-back and capacitive loading (Fig. 9). Asynchronous SAR logic enables fast SAR operation and removes the need for a fast SAR clock. To further reduce area, the CDAC reconfigures its capacitances to implement the two capacitors C_{SAM} and C_{SUM}

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TABLE IV
ADC SPECIFICATIONS

Specification	Value
Resolution	8 *bits*
Supply Voltage	1.2 *V*
Input Range	0 ~ (V_{REF+} - V_{REF-})
1 LSB	2.54 *mV*
Sampling Rate	40 *MHz*
Conversion Delay	20 *ns*
Steps	8 *sampling* + 1 *conversion*
ENOB	7.5 *bits*
Energy Efficiency	7.8 *pJ/conversion*

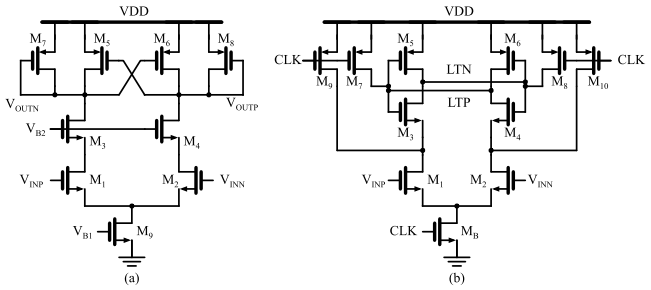


Fig. 10. (a) Preamplifier and (b) strong-ARM comparator.

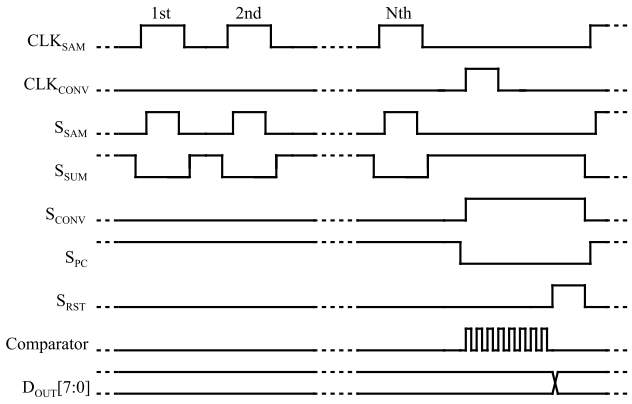


Fig. 11. BWMCS ADC timing.

in Fig. 8 for binary-weighted sampling. During sampling, the MSB capacitance of CDAC functions as the sampling cap, C_{SAM}, and the remainder of the CDAC capacitance functions as C_{SUM}. The CDAC unit capacitance is 1.4 fF, leading to a total CDAC capacitance of 360 fF, which provides a reasonable balance between kT/C noise and area. An extra switch S_{CONV} isolates C_{SUM} (i.e., $C_{\text{DAC}}[6:0]$) from the comparator's parasitic capacitance, guaranteeing precise charge-sharing and accurate weighting. A precharging switch S_{PC} closes when S_{CONV} opens to pre-charge the comparator input parasitic capacitance to a fixed voltage ($V_{\text{REF+}}$). After SAR conversion, a reset switch S_{RST} shorts the whole C_{DAC} capacitance to $V_{\text{REF}-}$ to clear the summed value. Fig. 11 shows a detailed timing diagram of the operation. Table IV summarizes the ADC specifications.

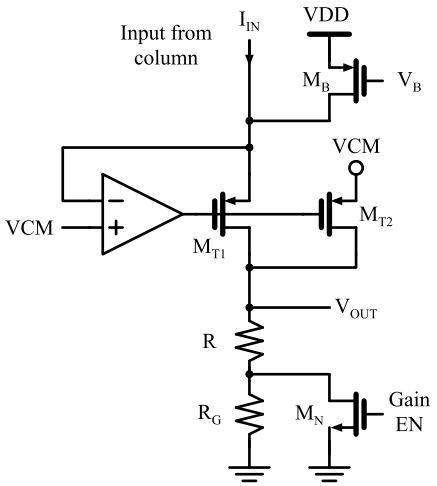


Fig. 12. g_m-boosted common-gate TIA with current amplification and variable gain.

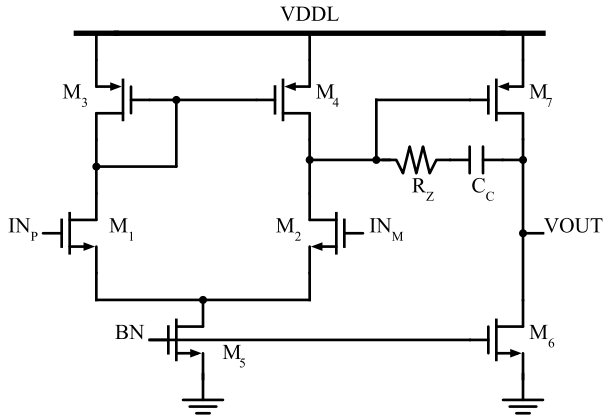


Fig. 13. Miller compensated TIA auxiliary amplifier.

B. G_m-Boosted Direct Drive TIA

A boosted-common-gate TIA terminates the crossbar column with a virtual ground, converts the column current to a voltage, and drives the ADC input. A modified boosted common-gate amplifier provides a high-quality virtual ground and is both area- and power-efficient. Moreover, a unique current amplification scheme enables the TIA to directly drive the ADC input capacitance.

Fig. 12 shows a depiction of the TIA. A g_m-boosted common-gate amplifier presents a low input resistance. A low TIA input impedance is essential for high-speed operation of a large ReRAM array. Since a BL column connects to many bit-cells, it accumulates a large parasitic capacitance from the bitcell 1T1R transistors. Unless the termination has a very low impedance, this capacitance degrades the transient response of the crossbar output current, thereby limiting crossbar accuracy and speed. g_m boosting reduces $1/g_m$ input resistance of the common-gate amplifier by the gain of the auxiliary amplifier (Fig. 13). The common-gate amplifier output current flows across a grounded resistor to form the ADC input voltage, V_{OUT}.

Fig. 13 shows the two-stage auxiliary amplifier, which drives the gate of the common-gate transistor to boost g_m.

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TABLE V
TIA SPECIFICATIONS

Specification	Single Read	MAC
Load	20 $k\Omega$	5 $k\Omega$
Gain	4	1
Input Range	0.3 - 3 μA	0 - 45 μA
Bleeding Current	30 μA	30 μA
Amplifier	x2	x2
Total Current	50.6 - 56 μA	50 - 140 μA

Miller compensation of the auxiliary amplifier sets the dominant pole of the TIA at the output of the first amplifier stage. The large range of BL current both during regular crossbar operation and while write-verifying a single bitcell presents a challenge for the stability of the TIA. When the BL current is low, the reduced common-gate transistor bias current increases the input resistance, and also greatly reduces the TIA input pole frequency. We add a bleed current from the current source M_B in parallel with the BL to alleviate this problem. The resulting increase in TIA output voltage due to this additional bias is easily canceled by adding a small dc offset to the ADC.

A current amplifier alleviates the gain-bandwidth tradeoff for the TIA output. A large TIA load resistance is beneficial because it increases the ADC input signal amplitude. A larger amplitude allows a larger sampling noise and therefore a smaller ADC sampling capacitance. However, this product of resistance and the ADC sampling capacitance sets the TIA bandwidth, limiting the speed of crossbar operation. An additional transistor (M_{T2}) has the same gate voltage and the same source bias voltage (i.e., V_{CM}) as the common-gate transistor (M_{T1}) and therefore acts as a current amplifier. Crucially, this allows a larger TIA output voltage without the need for a larger load resistance, thereby bypassing the *RC* constraint.

We also modify the TIA to support accurate characterization of bitcell conductance – this function is essential for the write-verification of filamentary ReRAM. Measuring an individual bitcell produces an μA-level current output, which is too small to be detected by the ADC. In bitcell measurement mode, we connect an additional series load resistance, under the control of GAIN EN, to boost the transimpedance and the resulting ADC input voltage. Table V shows the operating specifications of the TIA.

C. WL DAC

An accurate gate voltage for the 1T1R bitcells is vital for effective bitcell programming. The 9-bit WL DAC, which drives the 1T1R gates, consists of a binary-to-thermometer decoder, a 9-b segmented resistor array, and a unity gain buffer. To reduce the area, the 9-b segmented resistor array uses a 4-b thermometer resistor DAC array for higher bits and a 5-b R-$2R$ binary resistor DAC structure for lower bits, as shown in Fig. 14. An 8 kΩ unit resistance satisfies the linearity, speed, noise, and mismatch constraints. DAC switch sizes are weighted to reduce nonlinearity. The DAC output voltage range is from 0.5 to 2.9 V.

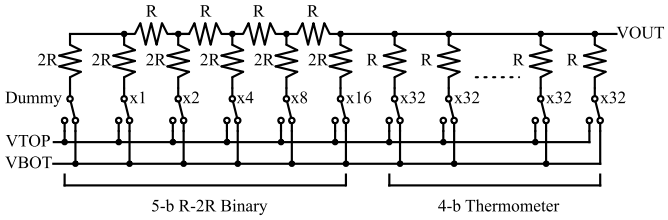


Fig. 14. WL 9-bit segmented resistor array DAC.

TABLE VI
WL DAC SPECIFICATION

Specification	Value
Resolution	9 *bits*
Supply Voltage	3.3 *V*
Top Reference Voltage	2.9 *V*
Bottom Reference Voltage	0.5 *V*
Output Voltage	0.5 - 2.9 *V*
1 LSB	4.69 *mV*
RDAC Current	1.36 *mA*
Buffer Current	312 μA
Bias Current	70 μA
Total Current	1.78 *mA*

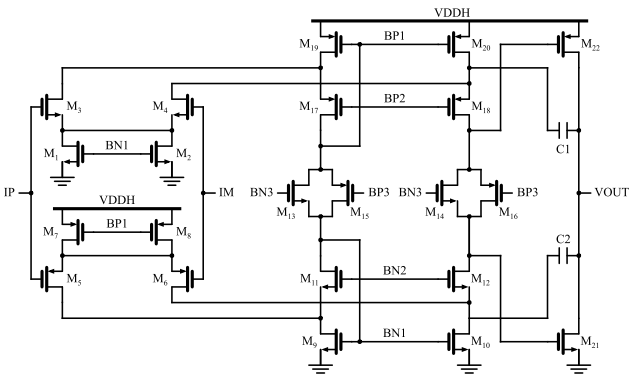


Fig. 15. Class AB rail-to-rail folded-cascode amplifier.

The DAC output voltage is buffered to drive the large WL capacitance including the 256 1T1R gates. A class-AB rail-to-rail folded-cascode amplifier [21] provides wide input and output ranges and saves power consumption in the unity-gain buffer (Fig. 15). The dc gain of the unity-gain buffer is 110 dB. The buffer settles in less than 50 ns for all conditions while loaded with 256 1T1R gates and related MUXes. Table VI summarizes the WL DAC specifications.

IV. SYSTEM CONTROL

A. Overview

The SoC architecture is composed of a RISC-V CPU, four CIM macros, a 32 KB instruction memory (IMEM), and a 1 MB data memory (DMEM), all connected to an AXI4 bus (Fig. 1). The RISC-V CPU [22] acts as the host for the CIM macros, facilitating the configuration and reconfiguration of the CIM macros, managing data movements to and from them, and monitoring their execution progress.

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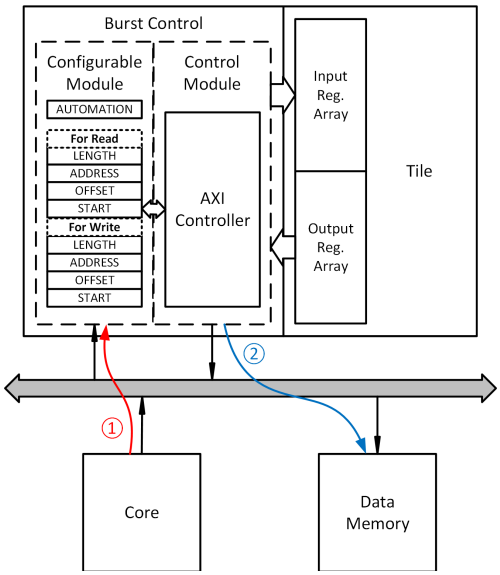


Fig. 16. Efficient data movement through burst control.

B. CIM Macro Control

Each CIM macro is equipped with a register interface that allows the host to exert precise control over the ReRAM cells. This control encompasses formation, setting, resetting, reading, and MAC-read operations, which are achieved by configuring the relevant registers within the CIM macro interface. However, while fine-grained control proves an advantage for testing and debugging purposes, it becomes impractical for the CPU to individually manage each ReRAM cell when executing extensive analog compute operations.

A dedicated controller embedded within each CIM macro operates independently of the CPU (Fig. 16). This allows the CPU to offload CIM timing and scheduling, resulting in significantly improved efficiency. The controller plays a crucial role in coordinating analog compute operations by providing control signals and enabling signals for clock gating. These signals are responsible for configuring crossbar hardware switches and driving the DACs and ADCs. Additionally, the controller supports parallel operation by simultaneously clocking out serialized data to the 1-b DACs from the DMA burst controller and operating the parallel samplers of BL ADCs.

C. Burst Mode for Efficient Data Transfer

The CIM timing controller incorporates a Burst Control mode, enabling accelerated data movement on the CIM macro without direct supervision from the host. The burst control includes two modules (Fig. 16): a configurable module and a control module. The configurable module is an AXI secondary module and receives operational details of the burst to initiate from the host. The operational details are stored in two sets of configurable registers, one for read and one for write. The control module is an autonomous AXI primary module that generates AXI accesses based on the values stored in the configurable registers. During the burst, the control module independently interacts with DMEM through the AXI bus without communicating with the host core.

The burst mode speeds up the data movement between the DMEM and the macro without the supervision of the host. To initiate a burst mode transaction, the host core first configures the configurable registers [Fig. 16 (1)]. The host core needs to specify the type of memory access by writing to either the set of registers for read or the set of registers for write. In either set of registers, multiple control fields need to be set in the corresponding registers: the total access length of the burst is set in LENGTH; the start address for memory read or write is set in ADDRESS; and the start offset index for writing or reading the input/output register array of the macro is set in OFFSET.

Once all configurations have been properly set, the host triggers a start signal through the AXI bus to a pseudo-register called START in the burst control. The START register does not store any value but serves as a signal for the control module in the burst control to initiate the burst transaction. Subsequently, the data movement between the DMEM and the macro is carried out exclusively by the control module in the burst control without the supervision of the host core [Fig. 16 (2)]. It is worth noting that each macro has its own dedicated bank in the DMEM to prevent conflicts with other macros during the burst transaction.

D. Decoupled Execution of Workload on Macros

The execution of workloads on macros is completely decoupled from the main process on the host core, resulting in higher efficiency in terms of communication latency and energy. A typical CNN inference workload involves three steps: 1) reading data from the DMEM to the macros; 2) computing the workload on the macros; and 3) writing data back to the DMEM. The burst mode transaction accelerates the data read before computation and the data write after computation. However, normally, the macro still needs to communicate with the host after each step is completed and waits for the host to initiate the next step. We take advantage of the fact that these communications between the macro and the host core are redundant. With minimal overhead, the burst control can assume the role of an execution controller, independently orchestrating the execution flow along with the data flow of the macro.

To activate the decoupled execution, the host sets the AUTOMATION register to 1. This requires setting both sets of configurable registers for read and write. The host specifies the total access length for data reading, the memory address from which to start reading data, and the starting offset index for placing data in the input register array of the macro. Similarly, the host specifies the data writing configurations, which inform the burst control on how to write results back to memory. Finally, the host initiates the execution by sending a start signal to the macro.

Upon receiving the start signal, the macro begins reading data from the DMEM via the AXI bus. The burst mode allows for quick access to the data. Once the data reading is complete, the burst control triggers the compute units in the macro for computation. After the computation is finished, the burst

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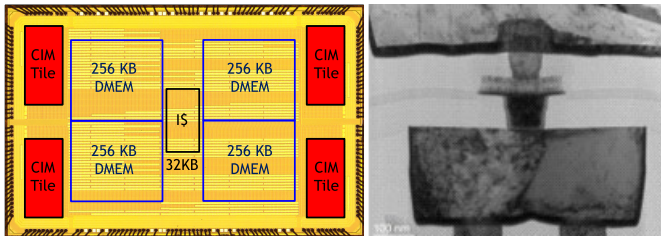


Fig. 17. Die photograph of (a) prototype SoC highlighting the four CIM macros and TEM image and (b) ReRAM deposited between the M4 and M5 metallization steps.

control initiates data writing to store the results back to the DMEM. When the entire workload is done, the burst control raises a done signal to notify the host.

While the workload is being executed, there is no need for communication between the host and macros. This allows the host to simultaneously initiate multiple macros to concurrently execute different workloads. This decoupled execution mode can be deactivated by the host, providing users the flexibility to exercise fine-grained control over the entire system. If the decoupled execution is deactivated, the macro communicates with the host to confirm completion after every step.

V. Prototype and Measurements

A. Overview

The prototype SoC is fabricated in 65 nm CMOS and occupies 16.5 mm^2 (Fig. 17). This IC includes four CIM macros, the RISC-V PULPino processor [22], 32 KB IMEM, and 1 MB DMEM. A single CIM macro comprises a 256 × 64 ReRAM crossbar array, 256 DACs, 32 TIAs, 32 ADCs, and crossbar periphery multiplexors. Each CIM macro occupies 0.27 mm^2. The input and output resolutions are both 8 bits. Operating from a 63 MHz system clock, the macro consumes 2.76 mW. The foundry-integrated multilevel 1T1R ReRAM crossbar array cells achieve up to 16 levels below 4 μA.

B. Packaging

A custom chip-on-board (COB) substrate accommodates the large pin count of the prototype and facilitates testing (Fig. 18). The ten-layer high-density interconnect (HDI) FR4 substrate is a fraction of the cost of a ceramic substrate yet provides significant advantages over conventional wire-bond packages such as QFN and QFP. The 3-mil substrate resolution allows fine-pitch pad placement, enabling short wire bond connections to the 176 IC pins. The substrate provides multiple ground planes and supports dense placement of decoupling capacitors. Edge-plated hard gold connections allow the entire substrate to function as a PLCC package. The custom substrate mounts in a standard 84-pin PLCC socket for easy evaluation of different samples.

C. CIM Macro Characterization

Full MAC characterization, including DL DAC, BL TIA, and BL ADC is performed in situ on a custom CIM macro

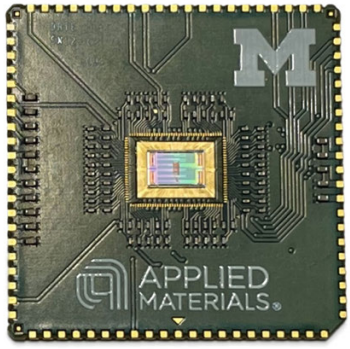


Fig. 18. Die on custom substrate.

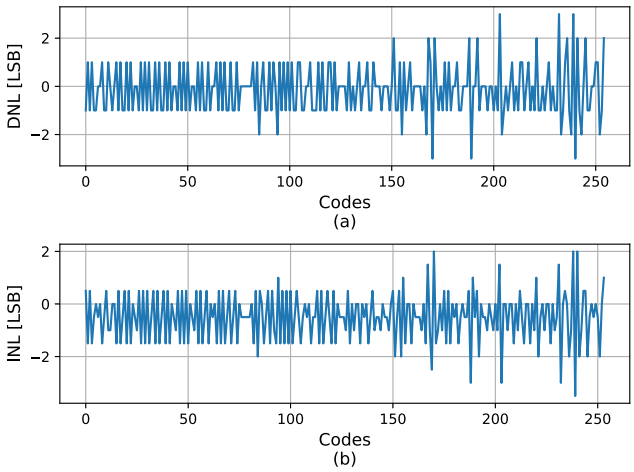


Fig. 19. Measured ADC (a) DNL and (b) INL.

with foundry polysilicon resistor devices replacing ReRAM bit-cells. We step through all 256 input DAC codes and collect the resulting ADC output for analysis. Fig. 19 shows the differential nonlinearity (DNL) and integral nonlinearity (INL) of the MAC calculation. The MAC SNR is 42.8 dB with an equivalent ENOB of 6.8 bits. The DNL measurement accuracy of this setup is limited by the 8-bit resolution of the CIM DAC, nevertheless, these measurements verify the excellent linearity of the ADC.

D. Multilevel ReRAM Device Programming

A write-verify algorithm combined with iterative step-pulse programming delivers 16 stable reset states below 4 μA. After the FORM process, the closed-loop double-sided programming algorithm tunes the bitcell conductance to the appropriate MLC state within a threshold of +/−500 nA. Fig. 20 shows the programming error for 3-bits per cell operation. Fig. 21 shows cumulative distribution function (cdf) plots of target programmed distributions and the effect of device retention. The plots show the distributions: (a) after programming, and (b) after one day. The states experience some drift but maintain excellent separation. The programming threshold, standard deviation error, and device retention are limitations of the experimental filamentary ReRAM devices, not the peripheral circuitry. Process-related details and further characterization of the ReRAM devices can be found in [14] and [15].

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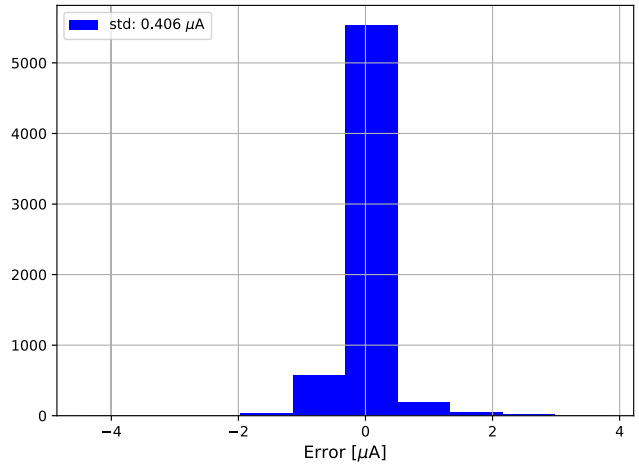


Fig. 20. Programming error for 3-bits per cell MLC ReRAM.

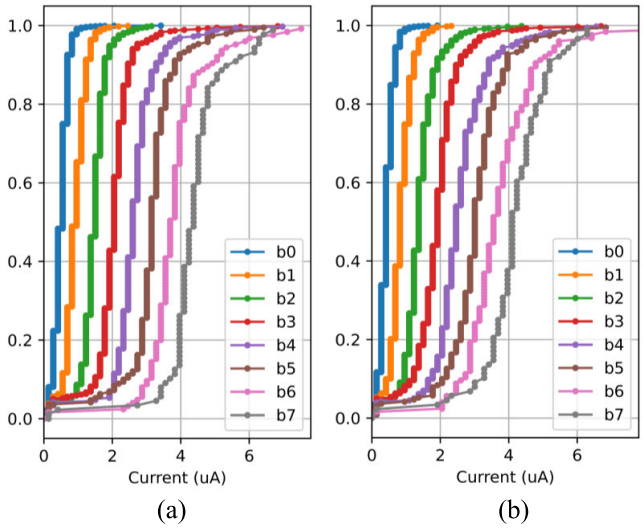


Fig. 21. Measured cdf of (a) initial ReRAM and (b) after one day.

E. DNN Implementation

We demonstrate the analog CIM capabilities of the prototype with an implementation of the LeNET 1 CNN architecture and Modified National Institute of Standards and Technology (MNIST) handwritten digit dataset [23], [24]. The network is composed of two convolutional layers and one fully connected (FC) layer connected to ten output classes. The CONV1 and CONV2 layers map to one CIM macro while the FC layer is mapped to a second CIM macro.

The weight values are optimized through QAT [25]. Quantization is applied in the forward pass during training, while the backpropagation is computed in high precision. In this way, the training infers the target quantized precision of the nonvolatile weights to recover classification accuracy. The classification accuracy of the trained model is 98.4%.

A dual array technique maps a vector of 4-bit signed weights onto adjacent 3-bit ReRAM columns [17]. The dual-array technique splits the trained LeNET1 filter into positive and negative arrays. Since two bitcells represent each weight, the effective weight resolution is 1-bit higher than that of the 3-bit ReRAM. The empty elements in each array are padded with zeros. Finally, the positive and negative arrays are flattened and assigned to adjacent crossbar columns. As mentioned in Section II, each column pair shares a TIA and ADC and, therefore, shares TIA/ADC offset and gain errors. The offset is common mode and is rejected.

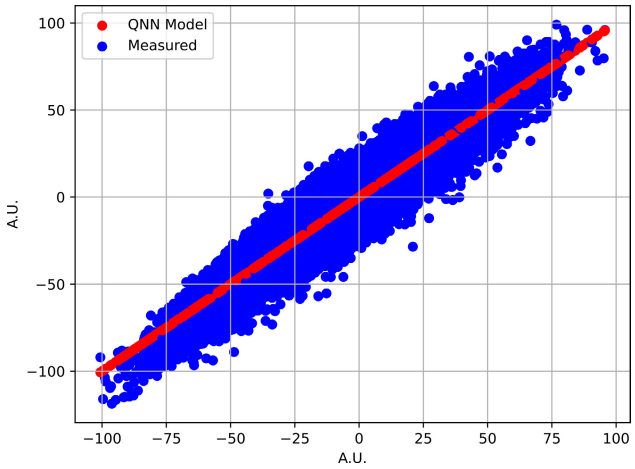


Fig. 22. Expected and measured output activations for all VMM operations on a batch of 128 images.

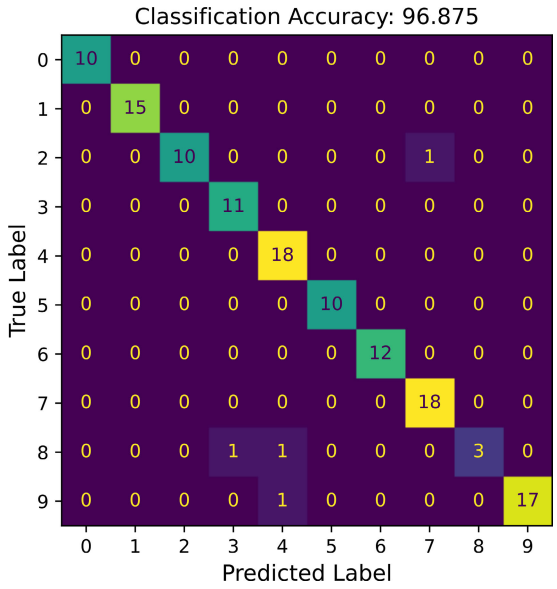


Fig. 23. Confusion matrix for a batch of 128 images from the MNIST validation set.

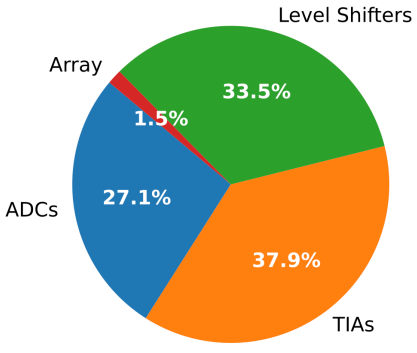


Fig. 24. CIM power consumption breakdown.

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TABLE VII
SYSTEM CHARACTERISTICS

Crossbar	
ReRAM Bitcell	1T1R filamentary
Array Dimension	256x64
Min. Bitcell Current	300 nA
Max. Bitcell Current	4 mA
Levels	Up to 16
Power	61 mW
Mixed signal Macro	
# of ADCs	32
ADC Resolution	8-bit
# of DACs	256
DL DAC	8-bit
Power	2.76 mW
System Performance	
System Clk	63 MHz
VMM/s	1.74 M
OP/s	57.1 GOPS
OPS/W	20.7 TOPS/W
RISC-V	100mW

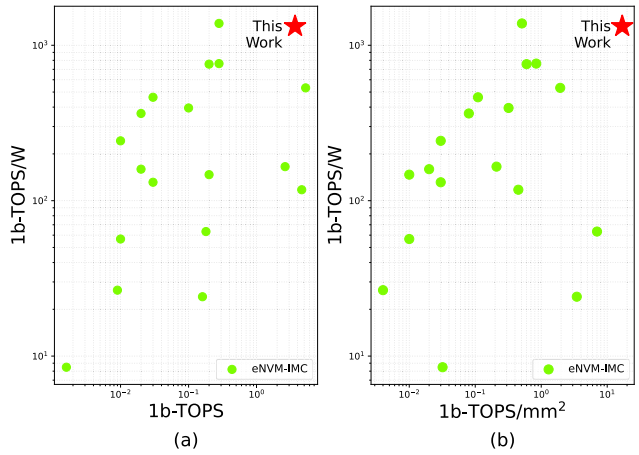


Fig. 25. (a) Energy-efficiency versus throughput and (b) energy-efficiency versus compute density for recent works [26], [27].

TABLE VIII
CONTRIBUTIONS AND BENEFITS

Contribution	Feature	Benefit
Throughput	ADC-assisted bit-serial scaling and processing	8-bit VMM in 8 cycles
Efficiency	Binary-weighting ADC Wide Bandwidth TIA	Energy-efficient VMM Free offset cancellation
Multi-level ReRAM	High/low voltage circuits ISPP with write-verify	16 level ReRAM
Compute Density	Pitch-matched circuitry Full parallelization	Maximized density and throughput

We use a write verify technique to accurately program the bitcells with 3-bit weights. A series of set and reset operations adjust the conductance. The write verify operation applies a read voltage to a single bitcell and determines the conductance from the column ADC output. We take advantage of the high-gain TIA mode to accurately measure the small current from a single bitcell during write verification.

TABLE IX
COMPARISON TABLE

	ISSCC '23 NTHU [12]	CICC '22 GaTech [13]	ESSCIRC '23 Stanford [16]	This Work
Technology	22nm	40nm	40nm	65nm
ReRAM Integration	AI-Edge Processor	Macro	Macro	Fully Integrated
ReRAM Bits	1 - 4	1	1 - 4	3
MLC Read/Write Circuitry	On-chip	On-chip	On-chip	On-chip
Bitcell Conductance	N/R	N/R	0 - 256 µS	0 - 13 µS
Parallel Write	No	No	Yes	No
Subarray Size	1Mb	256x256	512x64	256x64
Capacity	4 Mb	100Kb	3Mb	64Kb
Precision (I)	8	1	N/R	8
Precision (W)	8	1.58	N/R	4
Precision (O)	5	4	N/R	8
Column Sensing	5b CVH-ADC	4b ADC	Sense Amplifier	8b ADC
Active Rows	128	9	48	256
Operations Per ADC	128	N/A	48	256
Reported TOPS/W	51.4	118.4	N/R	20.7
Density TOPS/mm2	0.284	4.7	N/R	8.5

F. Measured DNN Performance

We perform inference on 128 images in the MNIST validation dataset to evaluate the performance of the CIM. Fig. 22 shows the expected output activations and calibrated output activations of the CONV1, CONV2, and FC layers for all VMMs. The red line is the expected result from the QNN model. The output activation errors (after partial summation) appear as gain and offset errors and are corrected using linear regression between the measured and expected output to learn the appropriate correction factors.

Fig. 23 shows a confusion matrix for a batch of 128 MNIST validation set images run on the prototype chip. The y-axis plots the true label provided by the MNIST dataset and the x-axis plots the output generated by the prototype.

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The network correctly classified 124 of 128 images, which corresponds to a 96.8% accuracy.

G. System Performance

We characterize the system performance during full array utilization. Specifically, we program a random 256 × 64 pattern into the array, continuously activate all rows and columns with random input vectors, and measure the average power consumption. Table VII summarizes the characteristics of the system and an individual CIM macro. Each CIM macro consumes 2.76 mW. As shown in Fig. 24, the level shifters, TIAs and ADCs each contribute approximately one-third of the power consumption. Fig. 25 compares the energy efficiency versus throughput and the energy efficiency versus compute density with recent works [26], [27].

We calculate the system throughput according to the following equation:

$$\frac{\text{Operations}}{\text{Sec}} = 2 * \frac{\text{VMMs}}{\text{sec}} * \text{CIM rows} * \text{CIM columns} \quad (2)$$

where the total throughput is scaled by the rows and columns of the crossbar array. System throughput is scaled by a factor of two, as both multiply and accumulate occur in a single step. The peak raw efficiency is calculated by normalizing the system throughput to the measured power consumption under full utilization, resulting in 20.7 TOPS/W (Table VII). When scaled to 1-bit by 1-bit MAC operation, the peak normalized efficiency is 662 TOPS/W. The compute density is 8.4 TOPS/mm^2.

VI. Conclusion

We present an SoC prototype comprised of four self-contained ReRAM-based CIM macros and a RISC-V host. Table VIII summarizes the main contributions and Table IX is a summary of recent CIM works. The measured raw and normalized peak efficiencies are 20.7 and 662 TOPS/W, respectively. The compute density is 8.4 TOPS/mm^2. The classification accuracy loss is 1.6% when compared to the quantized software model.

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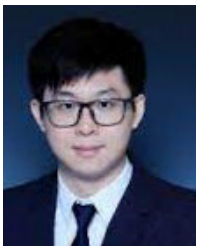
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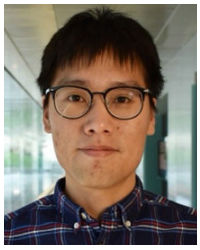
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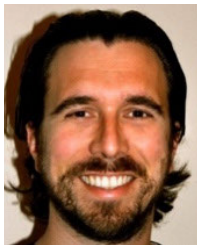
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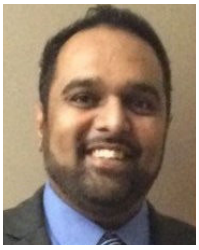
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